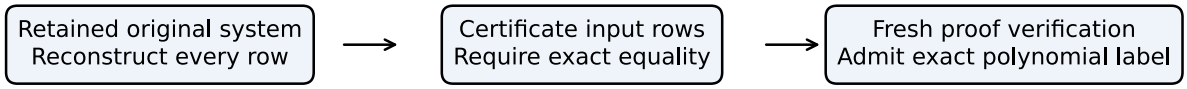
Original-system labels and an infinite bounded family

2026-11-09. Symbolic construction and software validation.



Whole dataset, group lifts and native cost remain separate

The new binding CLI reconstructs the original retained selector equations, requires identical certificate variables and literal rows, and re-runs the independent verifier. A valid certificate for another system is rejected. Exact labels are admitted only after fresh verification; whole-dataset, group-lift, native-cost and producer-authentication status remain false.

Forty-one software controls pass, including six new binding controls and the actual original-fixture label output. The broader workspace artifact search timed out after20seconds without a match; that incomplete discovery does not prove absence. No production case was replayed.

Infinite modular construction: For n=3*7^j, choose a Frobenius-stable trace-isotropic subspace L by selecting one component from each reciprocal factor pair, and use the trace-zero F8 plane S. With independent w,eta in S, set V=L+span(w,1+eta). Its dimension is(n+1)/2, restricted trace is nonzero and1 is outsideV.

On $E:y^2+xy=x^3+1$ with target(1,0), the full-vector-space Semaev system is UNSAT. The explicit affine separator lambda(t)=t+eta gives a literal original-system degree-three refutation. Consequently LFD<=3 for every member; exact degree is not asserted. The proof includes the Frobenius module construction, trace identities and degree-preserving lift from the trace slice.

This supplies an infinite low-degree UNSAT control family. It does not settle general fixed-m2 boundedness and does not transfer through arbitrary nonlinear rational-point selectors. The generator design is recorded but not implemented or executed.

Evidence: explicit-half-diagonal-affine-separator-family.md and independent reviews; attach-verified-lfd-source-review.md; bound-certificate-controls/. All new work is source-level proof or software validation. No scientific trials, native builds or cloud jobs occurred.